14/7/26

Practical - 4

Setting UP LOCAL AREA NETWORK (WAN)

Aim: To design, set up and configure a Local Area Network (LAN) connecting four computers through a network switch and demonstrate resource sharing across the network.

Equipment Required:

Equipment	Quantity	Specifications
personal computer	4	Ethernet (NIC) enabled
Network switch	1	8/16/24 ports
Ethernet cables	4	Straight through, RJ-45 connectors.

IP address configuration table.

Device	IP address	Subnet mask	Connected to
Pc0	10.1.1.1	255.0.0.0	Switch port Fa0/1
Pc1	10.1.1.2	255.0.0.0	Switch port Fa0/2
Pc2	10.1.1.3	255.0.0.0	Switch port Fa0/3
Pc3	10.1.1.4	255.0.0.0	Switch port Fa0/4
Switch	10.1.1.5	255.0.0.0	VLAN 1 interface

Implementation in Cisco packet tracer.

1) Place the Devices.

- Open Cisco packet tracer.
- From the device category panel, select End devices and drag 4 Pc's onto the workspace.
- Select switches and drag 1 switch.

2) Cable the devices.

- click Connections → select copper straight through cable.
- click each PC, then click the switch and select a available fast ethernet port to complete each connection.
- Repeat for all four PCs (Fa0/1 to Fa0/4) Green link lights confirm successful physical connections.

3) Assign IP addresses to PC's

- click a PC → Desktop tab → IP configuration.
- Enter the static IP address and subnet mask as per the IP table.

4) Configure the switch Management IP.

- click the switch → CLI tab.
- enable
  configure terminal
  interface vlan 1
  ip address 10.1.1.5 255.0.0.0
  shutdown
  exit.

5) verify connectivity using ping.

6) Demonstrate Data sharing (Server Shutdown)

- Add a server device (end devices → server PT)
- Assign it an IP address.
- Data transfer across LAN.

Result:

Thus LAN is setted up and connection is verified successfully.